

# Historical Shoreline Data

What I found in the Moore et al. 2013 working folder

Song

CEAMD Lab

May 6, 2026

## Two questions for today

- What's actually in the folder, and how usable is it?
- If wavelet analysis becomes a thing later, roughly what would it take?

*Just sharing what I found while poking around. Happy to be redirected.*

## What I did

- Walked the folder on the cluster, scanned all 147 `.shp` files
- Read the README, the GRL paper, and the supplemental methods
- Built an audit table for every file (CRS, bounds, year, geometry, source)
- Reprojected everything to a common CRS and plotted by year
- Read the existing DSAS `.xlsx` exports to recover the published numbers

*All in Python (geopandas / pandas / matplotlib); scripts available if helpful.*

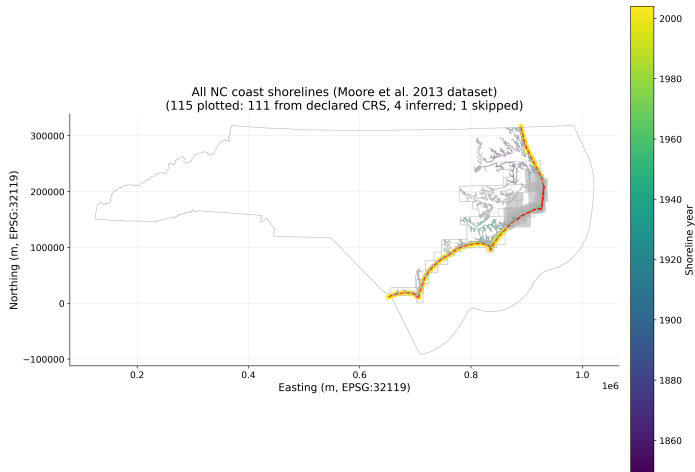
# The folder, in numbers

- **147** shapefiles in  
[abmurraylab/Shoreline Data from Moore et al 2013/](#)
- **116** are shorelines  
(*rest: baselines, transects, setback lines*)
- Years span **1849–2004**, 26 distinct dates
- Two focal regions: Cape Hatteras + Cape Lookout
- All maps onto Table S1 of Moore et al. 2013 GRL

## Top-level structure

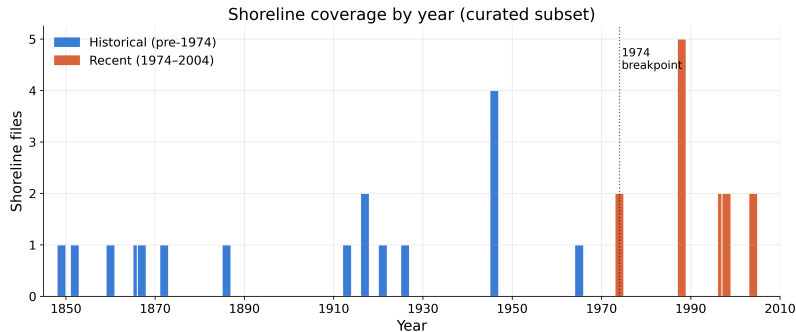
[shorelines/](#) raw shapefiles by source  
[Shoreline\\_Catalog/](#) Owen's catalog (= Table S1)  
[DSAS\\_cpHat/](#) ArcMap project + LRR results  
[DSAS\\_cpLO/](#) same, Cape Lookout  
[forDylanBrad.../](#) wave-climate workflow

## 115 shorelines, 1849–2004



*115 shoreline files plotted: 111 reprojected from their declared CRS, 4 assumed to be in NAD83 lat/lon from their coordinate bounds. 1 file skipped (empty geometry).*

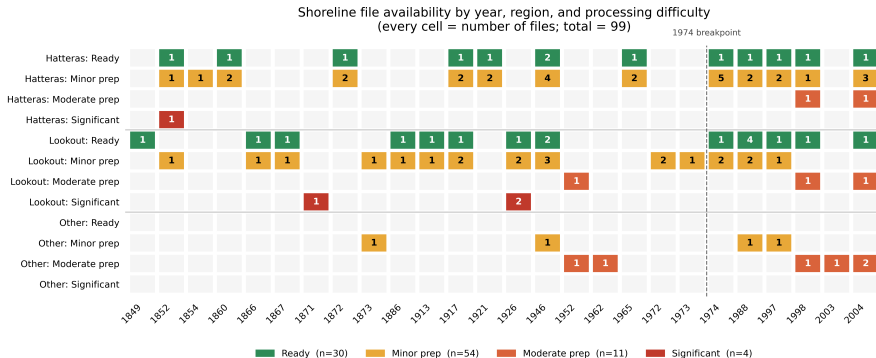
## Time coverage (Laura's 30 files)



- ~7 sampling years per cape pre-1974, ~5 post-1974
- Plenty for trend regression; far from enough for spectral methods

*This is the curated subset. The next slide shows what's in the rest of the folder.*

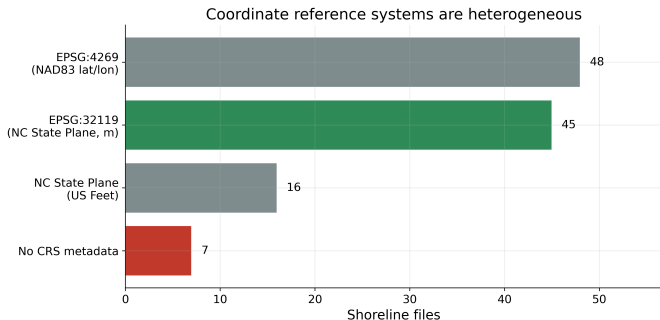
# What it would take to use the rest



- 30 ready-to-use (Laura's curated set)
- 54 minor prep — already in known CRS, just need a reproject step
- 15 more need either unit conversion (feet) or CRS guessing

*Numbers are file counts. Most files outside Laura's set already have a known CRS, so a reproject step is enough.*

# CRS is not uniform



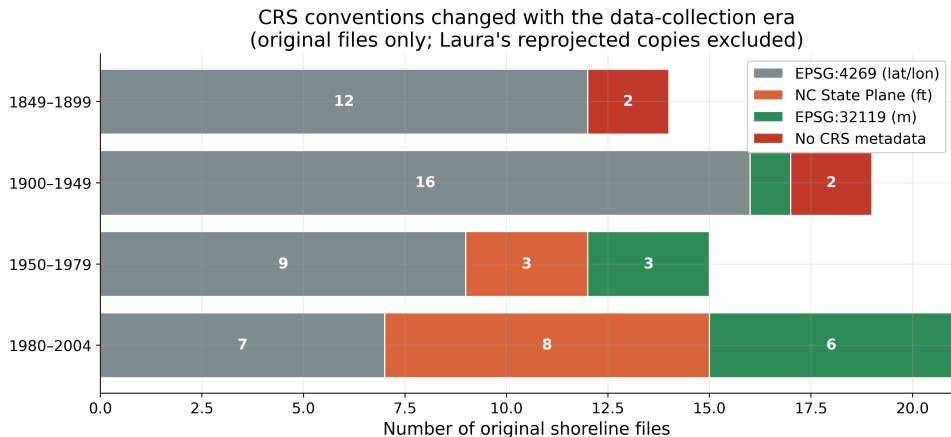
## What this means in practice:

- Files were prepared over many years, by different people
- Mixing them naively gives wrong distances and rates
- Some files have no .prj; CRS has to be *guessed*

**Good news:** Laura's reprojected\_shorelines/ subfolder has 32 files all in EPSG:32119.



# CRS conventions tracked the era of collection



*Older shorelines (NOAA T-sheets) are in geographic coordinates. NC State Plane (US Feet) shows up post-1950, with the digital era. The mess reflects when files were made, not how carefully.*

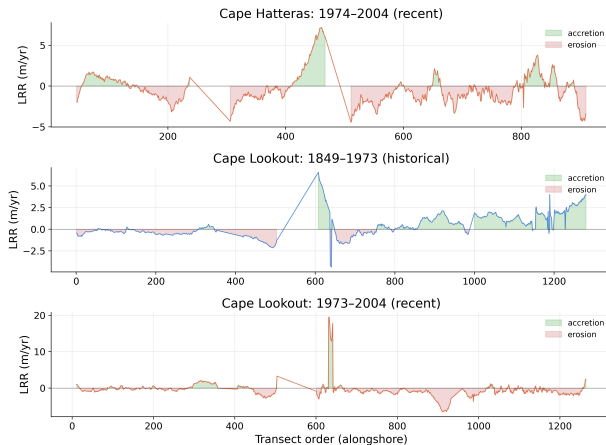
## What's already in the folder

- **Baselines** are drawn (standard\_baseline.shp, baseline\_tailored\*.shp)
- **Transects** are cast at 100 m intervals (trans\_standBL\*.shp)
- **LRR rates** are computed and exported to .xlsx:
  - Export52\_74stand.xlsx, Export74\_04stand.xlsx (Cape Hatteras)
  - Export\_1849\_1973.xlsx, Export\_t1973\_2004.xlsx (Cape Lookout)
- **Gaussian-smoothed values** are in RECALCaverageLRR.xlsx

*So Moore et al.'s analysis is sitting in the spreadsheets — DSAS doesn't need to run again.*

# Sanity check: Moore 2013 LRR, recovered from the exports

Moore et al. 2013 LRR results, recovered from existing exports



*Reading existing .xlsx in Python; same shape as Moore 2013 Figure 2. The  $\sim 20$  m/yr feature in the bottom panel is an inlet near Cape Lookout.*

## About wavelet analysis — initial thinking

- Current data:  $\sim 12$  points across 150 years per cape, irregularly spaced, time stamps not aligned across transects.
- What wavelet analysis would want: regular (e.g., quarterly or annual) shoreline positions at fixed transects over  $\geq 15$ –20 years.
- Where such data might live: NOAA Digital Coast lidar archive, USGS National Assessment of Shoreline Change updates, NC Beach Profile Monitoring (DCM).
- If we ever go there, the rough shape would be: extract a time series at each transect  $\rightarrow$  continuous wavelet transform (Morlet, e.g., [PyCWT](#))  $\rightarrow$  look for storm / seasonal / decadal modes.

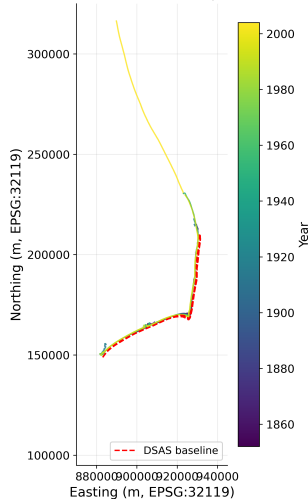
*This is a sketch, not a plan...*

## A few things I'm not sure about

- Is the goal to replicate Moore et al. 2013, or to extend it post-2004?
- Does Jose Antolinez's wave-climate workflow plug into this somewhere?
- Is anyone in the lab already in touch with NOAA / USGS / NC DCM about lidar shorelines?
- There are also .xlsx files I haven't read in detail (RECALCaverageLRR, smoothing parameters), and Jose's wave-climate workflow that I've only skimmed — worth digging into next?
- Anything in the folder I might have skipped over?

## Backup: Cape Hatteras detail

Cape Hatteras — 13 shorelines (curated subset)



## Backup: Cape Lookout detail

